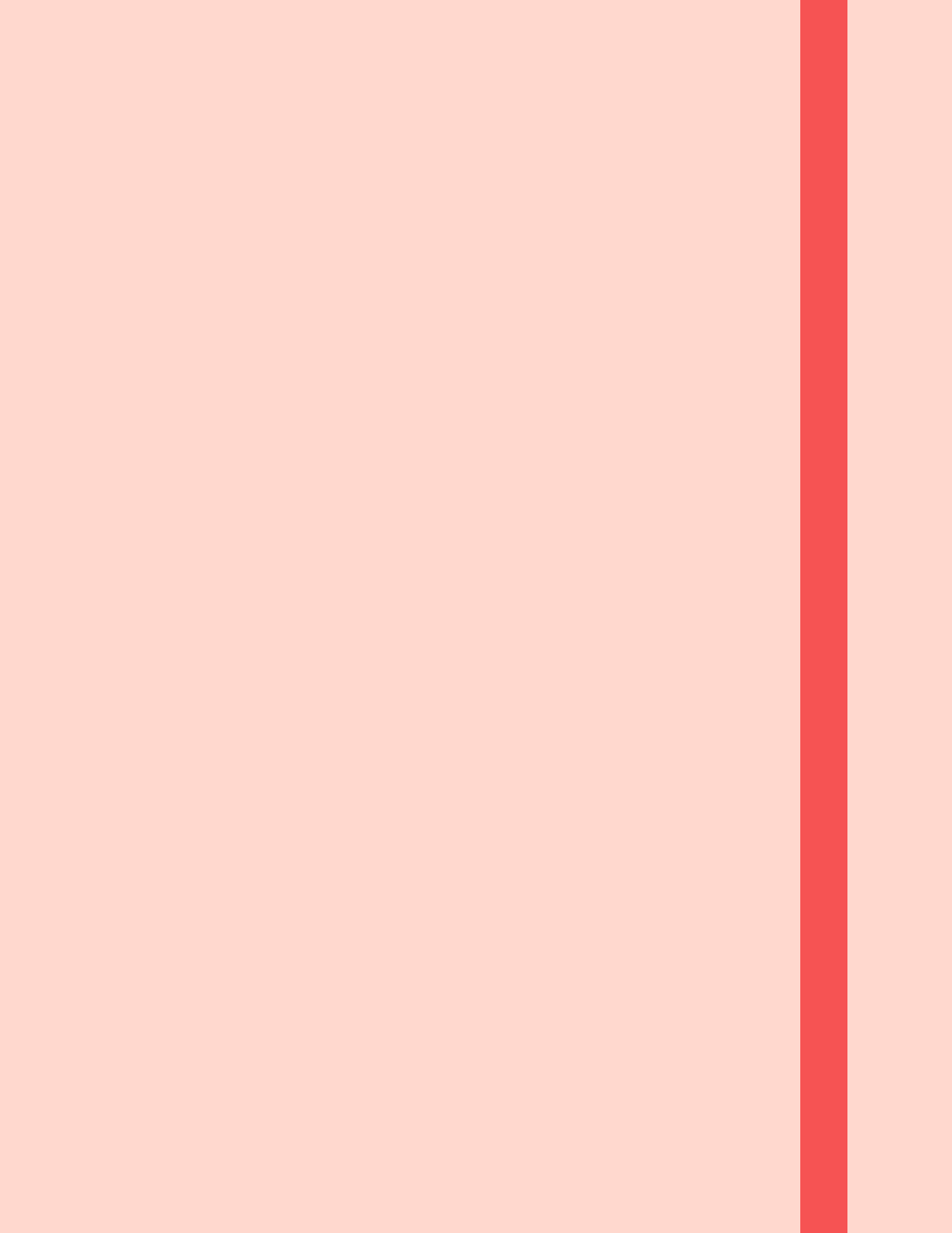




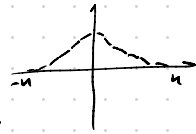
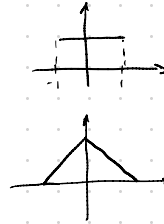
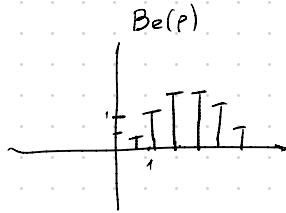
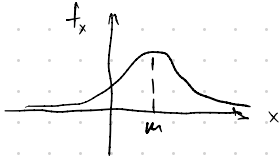
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Ивченко, Мерзев, Математическая статистика
Паннов, Паннов, Практикум по КС - задание

Многомерные гауссовские векторы

$$X \sim N(\mu, \sigma^2)$$

$$f_x(x) = \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



Опр. Нормальным (гауссовским) слуг. вектор назыв.

$$X: \Omega \rightarrow \mathbb{R}^n$$

$$f_x(x) = \frac{1}{\sqrt{(2\pi)^n \det K_x}} \exp\left\{-\frac{1}{2}(x-\mu_x)^T K_x^{-1}(x-\mu_x)\right\}$$

$\mu_x \in \mathbb{R}^n$ - вектор М.О.

$K_x \in \mathbb{R}^{n \times n}$ - ковариационная матрица

$$x \in \mathbb{R}^n$$

Если $X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \sim N(\mu_x, K_x)$, то $x_k \sim N(\mu_{x_k}, k_{x_k})$

Характеристическое уравнение

$$\text{Если } X \sim N(\mu_x, K_x)$$

$$\psi_x(j) = M[e^{j^T X}] = \exp\left\{j^T \mu_x - \frac{1}{2} j^T K_x j\right\}$$

Усл. Пусть $x \sim N(\mu_x, K_x)$, $y = AX + b$

$$\text{Тогда } y \sim N(\mu_y, K_y), \mu_y = A\mu_x + b$$

$$K_y = A K_x A^T$$

$$\textcircled{1} X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \sim N(\mu_x, K_x)$$

$$f_x(x) = C \cdot \exp\left\{-\frac{1}{2} Q(x_1, x_2)\right\}$$

$$Q(x_1, x_2) = 2x_1^2 + 3x_2^2 + 2x_1x_2 - 8x_1 - 14x_2 + 18$$

$$C = ? \quad \text{Значит переп. } (2x_1 - x_2) = ?$$

$$\begin{cases} \frac{\partial Q}{\partial x_1} = 4x_1 + 2x_2 - 8 \\ \frac{\partial Q}{\partial x_2} = 2x_1 + 6x_2 - 14 \end{cases} \begin{cases} 2x_1 + x_2 = 4 \\ x_1 + 3x_2 = 7 \end{cases} \begin{cases} x_1 = 1 \\ x_2 = 2 \end{cases}$$

$$u_x = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$K_x^{-1} = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} \quad K_x = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix}$$

$$x \sim N\left(\begin{pmatrix} 1 \\ 2 \end{pmatrix}; \begin{pmatrix} 3/5 & -1/5 \\ -1/5 & 2/5 \end{pmatrix}\right)$$

$$C = \frac{1}{\sqrt{(2\pi)^2 \cdot \frac{1}{5}}}$$

$$2x_1 - x_2 = \underset{1 \times 2}{(2-1)} \underset{2 \times 1}{\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}} + \underset{1}{0}$$

$$u_y = A u_x + b = (2 \ -1) \begin{pmatrix} 1 \\ 2 \end{pmatrix} + 0 = 0$$

$$K_y = \sigma_y^2 = (2 \ -1) \begin{pmatrix} 3/5 & -1/5 \\ -1/5 & 2/5 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \frac{18}{5}$$

$$y \sim N(0;$$

$$\textcircled{2} \quad \psi_x = (x_1, x_2) = \exp \{ (-x_1^2 + x_2^2) \}$$

$$P(|x_1 - x_2| \leq 6) = ?$$

$$\psi_x = \exp \left\{ i \underbrace{x^T u_x}_{=0} - \frac{1}{2} x^T K_x x \right\}$$

$$K_x = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \quad \overset{=0}{\Rightarrow} u_x = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$x \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}\right)$$

$$y = x_1 - x_2 = (1 \ -1) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + 0$$

$$u_y = 0$$

$$K_y = (1 \ -1) \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 4$$

$$y \sim N(0, 4)$$

$$P(|x_1 - x_2| \leq 6) = P(-6 \leq x_1 - x_2 \leq 6) =$$

$$= \Phi\left(\frac{6-0}{\sqrt{4}}\right) - \Phi\left(\frac{-6-0}{\sqrt{4}}\right) = \Phi(3) - \Phi(-3) = 2\Phi(3) = 2 \cdot 0.99865$$

Теорема о нормальской корреляции

$$\begin{pmatrix} x \\ y \end{pmatrix} \sim N\left(\begin{pmatrix} \mu_x \\ \mu_y \end{pmatrix}; \begin{pmatrix} K_x & K_{yx} \\ K_{xy} & K_y \end{pmatrix}\right) \quad K_{xy}^T = K_{yx}$$

$$M[|y - \hat{y}(x)|^2] \rightarrow \min$$

$$\Downarrow D[|y - \hat{y}(x)|^2]$$

Решение задачи (поиск функции)

самая маленькая ошибка

$$\hat{y}(x) = M[y|x]$$

$$\hat{y}(x) = \mu_{y|x} = \mu_y + K_{yx} \cdot K_x^{-1} (x - \mu_x)$$

$$K_{y|x} = K_y - K_{yx} K_x^{-1} K_{yx}^T$$

$$M[|y - M[y|x]|^2] = \text{tr } K_{y|x}$$

$$Law(y|x) \sim N(\mu_{y|x}; K_{y|x})$$

$$1. y = x + \varepsilon$$

$$X \sim N(\mu_x, \sigma_x^2)$$

Дит с.к. - оциска X по наб. y
 y по наб. X

$$\mu_{x|y} = ?$$

$$\varepsilon \sim N(\mu_\varepsilon, \sigma_\varepsilon^2)$$

$$\mu_{y|x}$$

$$X, \varepsilon - \text{н.з.}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} \sim N\left(\begin{pmatrix} \mu_x \\ \mu_y \end{pmatrix}; \begin{pmatrix} K_x & K_{yx} \\ K_{xy} & K_y \end{pmatrix}\right) = N\left(\begin{pmatrix} \mu_x \\ \mu_x + \mu_\varepsilon \end{pmatrix}; \begin{pmatrix} \sigma_x^2 & \sigma_x^2 \\ \sigma_x^2 & \sigma_x^2 + \sigma_\varepsilon^2 \end{pmatrix}\right)$$

$$\mu_{x|y} = \mu_x + K_{xy} K_y^{-1} (y - \mu_y) \Leftrightarrow$$

$$\text{cov}(X, y) = \text{cov}(X, X + \varepsilon) = \text{cov}(X, X) + \text{cov}(X, \varepsilon) = \sigma_x^2 + 0 = \sigma_x^2 = K_{xy}$$

$$\Leftrightarrow \mu_x + \sigma_x^2 (\sigma_x^2 + \sigma_\varepsilon^2)^{-1} (y - \mu_x - \mu_\varepsilon)$$

$$\mu_{y|x} = \mu_y + K_{yx} K_{xx}^{-1} (x - \mu_x) = \mu_x + \mu_\varepsilon + \sigma_x^2 \cdot \sigma_x^{-2} (x - \mu_x) = \mu_x + \mu_\varepsilon + x - \mu_x = x + \mu_\varepsilon$$

$$y_k = 2x + \varepsilon_k, \quad K = 1, 2$$

$$X \sim N(1, 4)$$

$$\varepsilon_1, \varepsilon_2, X - \text{н.з.}$$

$$\varepsilon_1, \varepsilon_2 \sim N(0, 1)$$

с.к. - оциска X по наб. y

$$y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

$$u_{x|y} = u_x + K_{xy} K_y^{-1} (y - u_y)$$

$$y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \sim N \left(\begin{pmatrix} u_{y_1} \\ u_{y_2} \end{pmatrix}; \begin{pmatrix} K_{y_1} & K_{y_1 y_2} \\ K_{y_2 y_1} & K_{y_2} \end{pmatrix} \right) = N \left(\begin{pmatrix} 2 \\ 2 \end{pmatrix}; \begin{pmatrix} 17 & 16 \\ 16 & 17 \end{pmatrix} \right)$$

$$\mathcal{D}[y_i] = \mathcal{D}[2x + \varepsilon_i] = \mathcal{D}[2x + \varepsilon_2] = 4\mathcal{D}[x] + \mathcal{D}[\varepsilon_i] = 17$$

$$M[y] - M[y|x] = 0 \sim K_{y|x}$$

$$\begin{aligned} \text{cov}(y_1, y_2) &= \text{cov}(2x + \varepsilon_1, 2x + \varepsilon_2) = \text{cov}(2x, 2x + \varepsilon_2) + \text{cov}(\varepsilon_1, 2x + \varepsilon_2) \\ &= \text{cov}(2x, 2x) + \text{cov}(2x, \varepsilon_2) + \text{cov}(\varepsilon_1, 2x) + \text{cov}(\varepsilon_1, \varepsilon_2) = 4\text{cov}(x, x) = 16 \end{aligned}$$

$$K_{xy} = (K_{xy_1} \quad K_{xy_2}) = (8 \quad 8)$$

$$\text{cov}(x, y_1) = \text{cov}(x, 2x + \varepsilon_1) = \text{cov}(x, 2x) + \text{cov}(x, \varepsilon_1) = 2\text{cov}(x, x) = 8$$

$$\begin{aligned} \text{cov}(x, y_2) &= 8 \\ u_{x|y} &= \mu_x + (8 \quad 8) \frac{1}{35} \begin{pmatrix} 17 & -16 \\ -16 & 17 \end{pmatrix} \begin{pmatrix} y_1 - 2 \\ y_2 - 2 \end{pmatrix} = \frac{8}{35} y_1 + \frac{8}{35} y_2 + \frac{1}{35} \end{aligned}$$

$$\begin{pmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \end{pmatrix} \sim N \left(\begin{pmatrix} 1 \\ 2 \\ 2 \\ 2 \end{pmatrix}; \begin{pmatrix} 2 & 1 & 0 & -1 \\ 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ -1 & 0 & -1 & 2 \end{pmatrix} \right)$$

$$\mathcal{L}(x|y) = N(u_{x|y}, K_{x|y})$$

$$\begin{aligned} u_{x|y} &= u_x + K_{xy} K_y^{-1} (y - u_y) = \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}^{-1} \begin{pmatrix} y_1 - 3 \\ y_2 - 4 \end{pmatrix} \\ &= \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} y_1 - 3 \\ y_2 - 4 \end{pmatrix} \end{aligned}$$

$$K_{x|y} = K_x - K_{xy} K_y^{-1} K_{xy}^T = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} - \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix} \leftarrow \text{D3}$$

Law $(x_1, y_1 | x_2, y_2)$